

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "House Finch"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

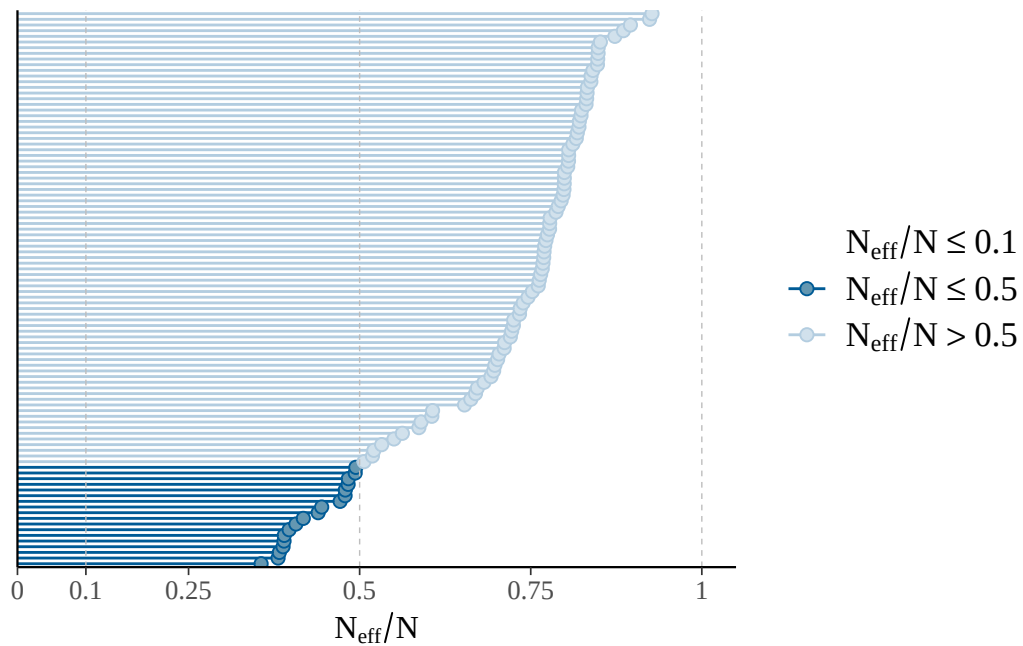
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3808	0.6864	0.9835	0.9802	1.1717	2.3678

```
[1] "n_eff ratio tail ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4408	0.6601	0.7634	0.7234	0.8128	0.9302

```
[1] "effective sample size in bulk and tail fine"
```

Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.





Post-warmup iteration

Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9992	0.9995	0.9999	1.0003	1.0005	1.0050

```
[1] "All R-hat values < 1.01"
```

Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_bio15_3yrs"          "b_occ_pr_spring_3yrs"
[3] "b_occ_ibio1_3yrsE2"        "b_occ_ibio2_3yrsE2"
[5] "b_occ_ibio7_3yrsE2"        "b_occ_ipr_autumn_3yrsE2"
[7] "b_occ_sum_annual_crops_3yrs" "b_occ_secdf_3yrs"
[9] "b_occ_pastr_3yrs"
```

```
[1] "b_colo_pr_winter"          "b_colo_ibio1E2"
[3] "b_colo_sum_annual_crops"   "b_colo_secdf"
[5] "b_colo_pastr"              "b_colo_urban"
[7] "b_colo_isum_annual_cropsE2" "b_colo_isecdfE2"
[9] "b_colo_ipastrE2"           "b_colo_lurbanE2"
```

```
[1] "b_ex_bio1"          "b_ex_bio3"          "b_ex_pr_spring"
[4] "b_ex_pr_summer"      "b_ex_ibio1E2"        "b_ex_ibio3E2"
[7] "b_ex_ipr_summerE2"   "b_ex_ipr_winterE2"   "b_ex_sum_annual_crops"
[10] "b_ex_secdf"          "b_ex_pastr"          "b_ex_urban"
[13] "b_ex_isecdfE2"       "b_ex_ipastrE2"
```

